Application Platform Delivery Foundations

Version 1, 2026-07-10

Developer Experience and GitOps Workflows

Introduction

Welcome to this quick course on the ***Foundations: Developer Experience and GitOps Workflows***. This course is the `first` in a series of `six` courses about **Application Platform Delivery**:

- Foundations: Developer Experience and GitOps Workflows (*This course*)
- *Module 2*

What's in this course?

- This course is designed to ……. implement its features, focusing on different user roles:

Enable learners to understand the principles of Developer Experience, Platform Engineering, and GitOps-based software delivery while preparing the environment and skills required for subsequent Software Factory implementation labs.

Duration: 2 hours

Contributors

The PTL team acknowledges the valuable contributions of the following Red Hat associates:

- .. (SME)
- .. (SME)
- Yogita Soni (Content Architect)
- Anna Swicegood (Learning Product Manager)

Objectives

On completing this course, you should be able to:

- …

Prerequisites

This course assumes that you have the following prior experience:

1. Red Hat OpenShift Container Platform (OCP) Technical Overview or Kubernetes Basics

Lab Environment

This classroom does **NOT** have Red Hat Connectivity Link (RHCL) pre-installed. You install and configure a RHCL in this course, and then continue for upcoming lab sections.

Click the RHDP link above, log in to RHDP, and click `Order` to request a new AWS with OpenShift Open Environment. On the order page for the catalog item, do the following:

- **Activity**: Select `Practice/Enablement`
- **Purpose**: Select `Learning about the Product`
- **Salesforce ID**: If you have an opportunity ID, enter it into this field, otherwise select `Project`, and then enter "Learning about RHCL" in the text field. You will see a warning message that a Salesforce ID is required, but you can safely ignore it.
- **Features**: Select `Enable Let's Encrypt`
- **Region**: Select the AWS region closest to your location
- **User count**: `1`
- **Control Plane Count**: `1`
- **OpenShift Version**: `4.16`
- **Control Plane Instance Type**: `m6a.4xlarge`
- Select the checkbox at the bottom of the page to confirm that lab costs will be charged to your cost center, and click `Order`.
- You can refer below image to confirm the lab settings.

[labenv] | *labenv.png*

> The lab environment will take an hour to provision. You can check the status and details of the OpenShift cluster from the `Services` page of RHDP. The classroom is available for 48 hours initially. You can extend the duration of classroom availability by clicking `Help > Get Technical Help` in RHDP, and then requesting an extension by opening a ServiceNow ticket with the RHDP team.

RHDP Portal Links

- RedHat Associates
- RedHat Partners

Understanding the Need for Platform Engineering

The Developer's Struggle

- Imagine you are a developer who has just finished building a *new* feature that customers have been eagerly waiting for. The code works perfectly on your machine, but now comes the difficult part—getting it into production.

▼ *Questions begin to pile up: Click here for a quick primer*

  - How do I onboard my application onto the platform?
  - Which tools, frameworks, and libraries should I use, and are they approved by the organization?
  - How do I deploy my code quickly and receive feedback faster?
  - What changes are required to modernize my application while continuing to deliver business value?
  - How can I ensure my application meets security and compliance requirements?
  - If I need new capabilities from the platform, how do I request them?

- Instead of focusing on delivering business value, developers often spend significant time navigating tooling decisions, understanding platform processes, modernizing applications, and coordinating with multiple teams. As complexity grows, innovation slows and delivery becomes more difficult.

The Platform Team's Struggle

- On the other side of the organization sits the platform team, responsible for building and operating the platforms that developers rely on every day.

- ▼ *Their challenges look different, but are equally demanding. Click here for a quick primer*
    - How do we increase adoption of the platform we’ve invested so much effort building?
    - How can we continue innovating while also maintaining and supporting existing services?
    - How do we manage developer requests for capabilities that cannot be immediately provided?
    - Why do developers see the platform as an obstacle rather than an enabler?
    - How do we reduce the never-ending backlog of support tickets and service requests?
    - How can we encourage developers to use the platform as intended without restricting productivity?
    - How do we empower developers with greater ownership while maintaining security and governance?
    - How do we measure whether the platform is actually delivering value to the organization?
- Platform teams often struggle to balance developer experience, operational excellence, security requirements, and business expectations—all at the same time.

DevOps Apocalypse

- For years, organizations embraced a "Do Everything" approach
    - ▼ *Click here for a quick overview of the challenges:*
        - Expecting developers to become experts in infrastructure, security, compliance, deployment pipelines, monitoring, and operations, while platform teams attempted to support every request manually.
        - As systems grew more complex, this model began to break down.
        - Developers became overwhelmed by cognitive overload.
        - Platform teams became bottlenecks buried under an ever-growing backlog of requests.

    - The result was slower delivery, increased frustration, and reduced innovation across the organization.

- Out of these challenges emerged Platform Engineering.
- Platform Engineering applies a product mindset to internal platforms, treating developers as customers and the platform itself as a product designed to solve their problems.
- By providing self-service capabilities, golden paths, and built-in security and governance, platform engineering enables developers to focus on what they do best—building applications that deliver business value.
- Instead of choosing between speed and control, organizations can achieve both.
- Platform Engineering accelerates software delivery through standardization, automation, and a relentless focus on developer experience.

What is DevEx and why is it a solution?

Developer Experience (DevEx) is the practice that makes Platform Engineering successful.

▼ *Click here to know: What is it?*

- Developer Experience (DevEx) is the sum of all factors that affect a developer's work throughout the Software Development Lifecycle (SDLC).
- It encompasses the tools, processes, platforms, environments, and organizational practices that influence how effectively developers can build, test, deploy, and operate software.
- The goal of DevEx is to reduce friction and enable developers to focus on delivering business value.

▼ *Click here to know: Why does it matter?*

- Platform Engineering succeeds when platforms are designed around the needs of developers.
- Developer feedback is critical and should flow continuously between developers and platform teams.
- This feedback helps identify friction points, prioritize improvements, and guide the evolution of the platform.
- By treating developers as customers, platform teams can create solutions that are easier to adopt and more effective to use.

▼ *Click here to know: Business Outcomes*

- Prioritizing Developer Experience makes it easier to develop, test, deploy, and operate software.
- Reduced friction leads to higher developer productivity and satisfaction.
- Faster feedback cycles enable teams to innovate more quickly and deliver value sooner.
- The result is improved software quality, faster innovation, and better business outcomes.

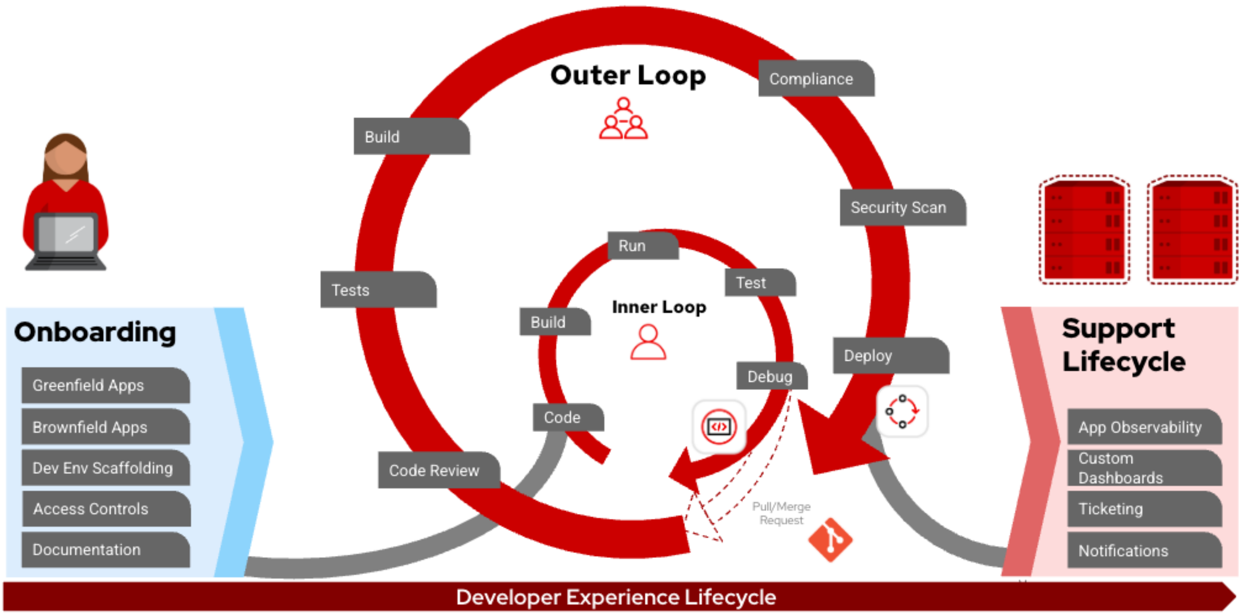
Benefits of a Developer-Centric Culture

- But you might be wondering why should companies care about measuring and improving developer productivity? *Well, it is simple!*

  - Companies that care and have a great developer centric culture report a massive boost in revenue - a 400% percent increase in revenue compared to companies that don’t have developer centric cultures.
- A Developer Centric culture delivers happiness!
  - Happier employees, in turn make companies more productive and result in that 4x revenue increase.
- If you get a chance, definitely read the [McKinsey article](). It is fascinating!
- Another big benefit of developer centric companies is employee retention. Happy, productive employees want to stay longer and that in turn drives more revenue.

Developer Experience Lifecycle

- Imagine a developer joining a platform for the first time.



- They begin with **onboarding**, gaining access to the tools, environments, and resources needed to get started.
- From there, they enter the **Inner Loop**—writing code, building, testing, and debugging applications.
- Once changes are ready, they move into the **Outer Loop**, where code is reviewed, validated, secured, and deployed through automated delivery pipelines.
- After deployment, applications enter the **Support Lifecycle**, where monitoring, observability, and operational processes help keep them running reliably.
- Together, these stages form the **Developer Experience Lifecycle**.
- Without a right integration between both loops (**Inner and Outer**), based on right processes supported by the Platform Teams, the Developer Experience will be impacted with low rates of productivity, poor performance, and low adoption rates of the platform created supposed to help or support them.
- This complexity must be addressed accordingly.

Accelerating Developer Productivity and Security

- *Imagine an organization that has invested in a modern application platform, yet developers still struggle with onboarding, deployment processes, tooling decisions, and slow feedback cycles. At the same time, platform teams are working hard to provide capabilities that developers may not be fully adopting or utilizing.*
- This is where a **Developer Experience Assessment** helps.
- Through a collaborative engagement, Red Hat Consulting works closely with platform teams and developers to understand how software is built, delivered, and operated across the organization.
- Together, they identify pain points, uncover opportunities for improvement, and align platform capabilities with developer needs.
- The outcome is a prioritized roadmap that helps improve:

    - Developer productivity
    - Strengthen security
    - Accelerate platform adoption.
- The roadmap also provides guidance for building key capabilities such as:
    - **An Internal Developer Portal (IDP)**
    - **A Trusted Software Supply Chain (TSSC)**
    - **Self-service developer experiences**
- *Success requires more than technology alone.*
- It depends on active collaboration, hands-on workshops, and open discussions where platform teams and developers work together to shape the future developer experience.

Check your work

Follow below steps in the lab to check your work

FIXME: Add the instructions to be followed in the lab environment to make sure the problem has been resolved. You may chose to remove this block if grading script is provided in the lab to check the work.

Check your work using the grading script

After you have resolved the issue and met the success criteria, login to `FIXME` system as `FIXME` user and run the grading script as shown here to check your work:

```
break-fix-script grade
```

Sample output

```
[root@classroom ~]# break-fix-script grade
Initiating break-fix-script with option grade
Grading.  Please wait.
Success.
COMPLETION CODE: <check your output for this value>
Completed break-fix-script with option grade successfully
```

FIXME: Remove the below block if no grading script is provided to check the work in the break-fix activity. In this case make sure you use the above block to provide step-by-step instructions learner can follow to check their work.

Submit the completion code

Skip this if not applicable for you.

- For RedHat internal audience: Submit that completion code here to receive a grade for this training
- For RedHat partners: Submit that completion code here to receive a grade for this training

FIXME: Change the links in the above lines to the link for appropriate audience specific LMS where learner can provide completion code. Remove this block if submitting completion is not applicable.

Guided solution

This section presents a guided solution to the break-fix activity.

Use this section to:

Learn one possible path to resolving the issue.

Get some tips for investigating and resolving similar issues.

If you're still stuck after exhausting your expertise and resources on the break-fix activity:

Use this solution to complete the activity. Check your work when you finish to receive a grade for this activity.

The steps detailed in these solution pages are optional. They provide additional guidance for those struggling with the break-fix activity. We encourage you to only use these pages if you are completely stuck in solving the issue

Guided solution (page 1 of 2)

FIXME: Provide step-by-step instructions here to resolve the issue.

Guided solution (page 2 of 2)

FIXME: Provide step-by-step instructions here to resolve the issue.